Teacher Wu Kuan | See It at a Glance #17 — $\sin^2\theta + \cos^2\theta = 1$ from One Unit Circle

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1. The trap question

Why is $\sin^2\theta + \cos^2\theta = 1$? If $\sin\theta = 3/5$, what is $\cos\theta$?

2. The method

- A unit-circle point is $(\cos\theta, \sin\theta)$, distance 1 from the origin.
- Pythagoras: $\cos^2\theta + \sin^2\theta = 1$.
- $\sin\theta = 3/5 \rightarrow \cos^2\theta = 16/25 \rightarrow \cos\theta = 4/5$ (acute angle).

3. Parent check method

Ask your child to draw the right triangle inside the unit circle and label the legs.

Practice

1. θ acute, $\sin\theta = 5/13$. Find $\cos\theta$.
2. $\cos\theta = 0.6$. Find $\sin\theta$.
3. Find $\sin^2 20° + \cos^2 20°$.
4. $\sin\theta = 3/5$ (acute). Find $\tan\theta$.
5. $\sin\theta \cos\theta = 1/4$. Find $(\sin\theta + \cos\theta)^2$.

Answers

1. 12/13
2. 0.8
3. 1
4. 3/4 ($\cos\theta = 4/5$)
5. 3/2 ($= 1 + 2 \times 1/4$)

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